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CHEAT SHEET - LINEAR BOUNDED AUTOMATA (LBA)

(An LBA is a TM whose tape is bounded by the input length. So all solutions use marking, scanning, and reuse of input cells only)

GROUP 1: $a^n b^n$ and Fixed Ratio Matching (LBA Version)

Questions

- $L = \{ a^n b^n : n > 0 \}$
- $L = \{ a^{2n} b^n : n > 0 \}$
- $L = \{ a^{3n} b^n : n > 0 \}$
- $L = \{ a^n b^{2n} : n > 0 \}$
- $L = \{ a^n b^{3n} : n > 0 \}$

Key Idea

1. Scan and mark one a
 2. Move right and mark k bs
 3. Return to leftmost unmarked a
 4. Accept if no unmarked a or b remains
- LBA restriction satisfied: only reuse input cells
Marking symbols: $a \rightarrow X, b \rightarrow Y$

GROUP 2: $a^n b^n$ with Extra Blocks

Questions

- $L = \{ a^n b^n c^m : n, m > 0 \}$
- $L = \{ a^n b^m c^m : n, m > 0 \}$
- $L = \{ a^m b^n c^m : n, m > 0 \}$
- $L = \{ a^n b^n c^n : n > 0 \}$

Key Idea: Partial matching inside bounded tape 1. First verify the block order ($a^* b^* c^*$)

2. Match only the required blocks
3. Ignore the remaining block
4. Accept when all required matches succeed

GROUP 3: Equality & Inequality of Symbol Counts (Two Symbols) Questions

- $L = \{ w : n_a(w) = n_b(w) \}$
- $L = \{ w : n_a(w) > n_b(w) \}$
- $L = \{ w : n_a(w) < n_b(w) \}$
- $L = \{ w : n_a(w) \neq n_b(w) \}$

Key Idea

1. Repeatedly cancel one a with one b
2. Mark both symbols
3. Check which symbol remains at the end
4. Leftover symbols decide inequality

GROUP 4: Equality & Inequality (Three Symbols)

Questions

- $L = \{ x \mid N_a(x) = N_b(x) = N_c(x) \}$
- $L = \{ x \mid N_a(x) < N_b(x) \}$
- $L = \{ x \mid N_a(x) < N_c(x) \}$
- $L = \{ x \mid N_b(x) < N_c(x) \}$
- $L = \{ x \mid N_a(x) < N_b(x) < N_c(x) \}$

Key Idea

1. Cancel symbols pairwise (a–b, b–c, etc.)
2. Compare which symbols remain
3. Accept based on condition

GROUP 5: Index Comparison Languages (i < j type)

Questions

- $L = \{ a^i b^j c^k : i < j \}$
- $L = \{ a^i b^j c^k : i < k \}$
- $L = \{ a^i b^j c^k : j < k \}$
- $L = \{ a^i b^j c^k : i < j \text{ and } i < k \}$

Key Idea: Length comparison using marking

1. Cross off one symbol from smaller block
2. Try to match symbols in larger block
3. If larger block still has unmarked symbols → accept

GROUP 6: Palindrome & Mirror Languages

Questions

- $L = \{ w \mid w \text{ is a palindrome} \}$
- $L = \{ w w^r \mid w \in \{a, b\}^* \}$
- $L = \{ w c w^r \mid w \in \{a, b\}^* \}$

Key Idea

1. Mark leftmost symbol
2. Move to rightmost unmarked symbol
3. Compare symbols
4. Repeat until center is reached

GROUP 7: Duplication & Repetition Languages

Questions

- $L = \{ w w \mid w \in \{a, b\}^* \}$
- $L = \{ w w \mid w \in \{a, b, c\}^* \}$
- $L = \{ x x x \mid x \in \{a, b\}^* \}$
- $L = \{ a^i b^j c^i d^j : i \geq 0 \}$

Key Idea: Comparison only, no copying

1. Mark symbols of the first block
2. Compare with corresponding symbols in next block(s)
3. Accept if all symbols match

GROUP 8: Separator-Based Matching

Questions

- $L = \{ w c w \mid w \in \{a, b\}^* \}$
- $L = \{ w c w^r \mid w \in \{a, b\}^* \}$

Key Idea

1. Treat separator as middle boundary
2. Match left and right parts symbol by symbol
3. Accept if all symbols match

GROUP 9: Multiple Equality Constraints

Questions

- $L = \{ a^i b^j c^k d^l \mid i = j \text{ and } k = l \}$
- $L = \{ a^i b^j c^i d^j \mid i \geq 0 \}$

Key Idea: Multi-pass marking approach

1. Phase 1: Match a with b
2. Phase 2: Match c with d
3. Accept only if both phases succeed



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CHEAT SHEET - LANGUAGE CLASSES (Recursive, Recursively Enumerable, Regular, CFL, CSL):

Regular $\subseteq$ CFL $\subseteq$ CSL $\subseteq$ Recursive $\subseteq$ RE

GROUP 1: Relationship Between Recursive and Recursively Enumerable (RE) Languages

Questions

- Prove that every recursive language is recursively enumerable.
- If L is a recursively enumerable language and its complement is also recursively enumerable, then prove that L is recursive.
- Show that if L is a recursive language, then its complement is also recursive.

Key Idea / Proof Template

- Recursive language $\Rightarrow$ TM halts on all inputs
- RE language $\Rightarrow$ TM may not halt on non-members
- If both L and $\bar{L}$ are RE, run both TMs in parallel
- One must accept $\Rightarrow$ Decider exists

Core concept: Decidability vs Semi-decidability

GROUP 2: Closure Properties of Recursive Languages

Questions
If L_1 and L_2 are recursive languages, show that:

$L_1 \cup L_2$ is recursive

$L_1 \cap L_2$ is recursive

Key Idea / Proof Template

- Recursive languages have deciders
- Combine deciders using: OR logic for union, AND logic for intersection
- Since both halt, combined TM halts

Important note: Recursive languages are closed under complement, union, intersection

GROUP 3: Closure Properties of Recursively Enumerable Languages

Questions

If L_1 and L_2 are recursively enumerable languages, show that:

- $L_1 \cup L_2$ is recursively enumerable
- $L_1 \cap L_2$ is recursively enumerable
- $L_1 - L_2$ is recursively enumerable

Key Idea / Proof Template

- RE languages have recognizers, not deciders
- Use parallel simulation (dovetailing)
- Union: accept if either recognizer accepts
- Intersection: accept if both accept
- Difference: accept if L_1 accepts and L_2 does not accept

Note: RE languages are not closed under complement

GROUP 4: Regular Languages and Higher Language Classes

Questions

- Prove that every regular language is recursively enumerable.
- Prove that every regular language is context sensitive.

Key Idea / Proof Template

- Regular language $\Rightarrow$ DFA exists
- DFA can be simulated by: Linear Bounded Automaton (LBA), Turing Machine
- Hence belongs to higher classes

Hierarchy idea: Regular $\subseteq$ CSL $\subseteq$ RE

GROUP 5: Context-Free and Context-Sensitive Languages

Questions

- Prove that every context-free language is context sensitive.
- Prove that every context sensitive language is recursively enumerable.

Key Idea / Proof Template

- CFL $\Rightarrow$ PDA exists
- PDA can be simulated by LBA, LBA is a restricted TM
- Therefore CSL $\subseteq$ RE

GROUP 6: Closure Properties of Context Sensitive Languages

Questions

If L_1 and L_2 are context sensitive languages, prove that:

$L_1 \cup L_2$ is context sensitive
 $L_1 \cap L_2$ is context sensitive
 $L_1 - L_2$ is context sensitive

Key Idea / Proof Template

CSLs are accepted by Linear Bounded Automata

LBAs are closed under:

Union
Intersection
Complement
Difference follows from intersection + complement

Important contrast: CFLs are not closed under intersection or complement

GROUP 7: Language Hierarchy & Containment Proofs

- Show the Chomsky hierarchy of languages.
- Prove that Recursive $\subset$ Recursively Enumerable.
- Give an example of a recursively enumerable but not recursive language.
- Explain why the Halting Problem is recursively enumerable but not recursive.

Key Idea

Use known undecidable problems

Show existence of recognizer but no decider



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CHEAT SHEET - TURING MACHINE

GROUP 1: $a^n b^n$ and Fixed Ratio Matching Languages

Questions

$L = \{ a^n b^n : n > 0 \}$

$L = \{ a^{2n} b^n : n > 0 \}$

$L = \{ a^{3n} b^n : n > 0 \}$

$L = \{ a^n b^{2n} : n > 0 \}$

$L = \{ a^n b^{3n} : n > 0 \}$

Key Idea / Template

1. Repeatedly mark one a
2. Move right and mark k bs
3. Return to left and repeat
4. Accept if no unmarked a or b remains

Used technique: Symbol marking & ratio matching

GROUP 2: $a^n b^n$ with Extra Symbols

Questions

$$L = \{ a^n b^n c^m : n, m > 0 \}$$

$$L = \{ a^n b^m c^m : n, m > 0 \}$$

$$L = \{ a^m b^n c^m : n, m > 0 \}$$

$$L = \{ a^n b^n c^n : n > 0 \}$$

Key Idea / Template

1. First verify block structure ($a^* b^* c^*$)
2. Match only the required blocks
3. Ignore unrelated symbols
4. Accept when matching succeeds

Used technique: Partial matching + block verification

GROUP 3: Equality of Symbol Counts (Two Symbols)

Questions

$$L = \{ w : n_a(w) = n_b(w) \}$$

$$L = \{ w : n_a(w) > n_b(w) \}$$

$$L = \{ w : n_a(w) < n_b(w) \}$$

$$L = \{ w : n_a(w) \neq n_b(w) \}$$

Key Idea / Template

1. Cancel one a with one b
2. Repeat until one symbol finishes
3. Decide acceptance based on leftover symbols

Used technique: Pairwise cancellation

GROUP 4: Equality & Inequality (Three Symbols)

Questions

$$L = \{ x \mid N_a(x) = N_b(x) = N_c(x) \}$$

$$L = \{ x \mid N_a(x) < N_b(x) \}$$

$$L = \{ x \mid N_a(x) < N_c(x) \}$$

$$L = \{ x \mid N_b(x) < N_c(x) \}$$

$$L = \{ x \mid N_a(x) < N_b(x) < N_c(x) \}$$

Key Idea / Template

1. Repeatedly cancel required symbol pairs
2. Check remaining symbols to decide inequalities

Used technique: Multi-symbol comparison

GROUP 5: Index Comparison Languages ($i < j$ type)

Questions

$$L = \{ a^i b^j c^k : i < j \}$$

$$L = \{ a^i b^j c^k : i < k \}$$

$$L = \{ a^i b^j c^k : j < k \}$$

$$L = \{ a^i b^j c^k : i < j \text{ and } i < k \}$$

Key Idea / Template

1. Cross off one symbol from the smaller block
2. Try to match with the larger block
3. If the larger block still has symbols → accept

Used technique: Block length comparison

GROUP 6: Palindrome & Mirror Languages

Questions

$$L = \{ w \mid w \text{ is a palindrome} \}$$

$$L = \{ w w^r \mid w \in \{a, b\}^* \}$$

$$L = \{ w c w^r \mid w \in \{a, b\}^* \}$$

Key Idea / Template

1. Mark leftmost symbol
2. Move to rightmost unmarked symbol
3. Compare symbols
4. Repeat until middle is reached

Used technique: Two-end scanning

GROUP 7: Duplication & Copy Languages

Questions

$$L = \{ w w \mid w \in \{a, b\}^* \}$$

$$L = \{ w w w \mid w \in \{a, b\}^* \}$$

$$L = \{ x x x \mid x \in \{a, b\}^* \}$$

$$L = \{ a^i b^j c^i d^j \mid i \geq 0 \}$$

Key Idea / Template

1. Mark symbols of first block
2. Copy/compare with remaining blocks
3. Accept if all blocks match

Used technique: Mark–copy–compare

GROUP 8: Separator-Based Matching

Questions

$$L = \{ w c w \mid w \in \{a, b\}^* \}$$

$$L = \{ w c w^r \mid w \in \{a, b\}^* \}$$

Key Idea / Template

1. Use separator as midpoint
2. Match left and right parts symbol by symbol

Used technique: Separator-guided matching

GROUP 9: String Manipulation / TM as Function Computer

Questions

String copy operation
String reverse operation
 $f(x, y) = xy$
 $f(x, y) = |x - y|$ (unary)
 $f(x) = 2x$

Key Idea / Template

1. Use delimiter symbols
2. Use extra tape space
3. Halt with output on tape

Used technique: TM as transducer

GROUP 10: Advanced Equality with Multiple Constraints Questions

$L = \{ a^i b^j c^k d^l \mid i = j \text{ and } k = l \}$

Key Idea / Template

1. First match a with b
2. Then match c with d
3. Accept only if both succeed

Used technique: Multi-phase matching



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Homework: CSG

Write CSG to generate the language $L = \{ a^n b^n : n > 0 \}$.

Hint:

Grammar: $G = (\{S, B\}, \{a, b\}, P, S)$

Production Rules:

$S \rightarrow aSB$

$S \rightarrow ab$

$bB \rightarrow bb$

$aB \rightarrow ab$

Write CSG to generate the language $L = \{ a^{2n} b^n : n > 0 \}$.

Hint:

Grammar: $G = (\{S, B\}, \{a, b\}, P, S)$

Productions

1. $S \rightarrow aaSB$
2. $S \rightarrow aaB$
3. $aB \rightarrow ab$
4. $bB \rightarrow bb$

Write CSG to generate the language $L = \{ a^n b^{2n} : n > 0 \}$.

Hint:

Grammar: $G = (\{S, B, C\}, \{a, b\}, P, S)$

Productions

1. $S \rightarrow aSBC$
2. $S \rightarrow aBC$
3. $aB \rightarrow ab$
4. $bB \rightarrow bb$
5. $bC \rightarrow bb$

Write CSG to generate the language $L = \{ a^{3n} b^n : n > 0 \}$.

Hint:

Grammar: $G = (\{S, B\}, \{a, b\}, P, S)$

Productions

1. $S \rightarrow aaaSB$
2. $S \rightarrow aaaB$
3. $aB \rightarrow ab$
4. $bB \rightarrow bb$

Write CSG to generate the language $L = \{ a^n b^{3n} : n > 0 \}$.

Hint:

$G = (\{S, B\}, \{a, b\}, P, S)$

Productions

1. $S \rightarrow aSBBB$
2. $S \rightarrow aBBB$
3. $aB \rightarrow ab$
4. $bB \rightarrow bb$

Write CSG to generate the language $L = \{ a^n b^n c^m : n, m > 0 \}$.

Hint:

$G = (\{S, B, C\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow aSB$
2. $S \rightarrow ab$
3. $aB \rightarrow ab$
4. $bB \rightarrow bb$
5. $S \rightarrow SC$
6. $C \rightarrow c$

Write CSG to generate the language $L = \{ a^n b^m c^m : n, m > 0 \}$.

Hint:

$G = (\{S, B\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow aS$
2. $S \rightarrow B$
3. $B \rightarrow bBc$
4. $B \rightarrow bc$

Write CSG to generate the language $L = \{ a^m b^n c^m : n, m > 0 \}$.

Hint:

$G = (\{S, T\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow aSc$

2. $S \rightarrow T$

3. $T \rightarrow bT$

4. $T \rightarrow b$

Write CSG to generate the language $L = \{ a^n b^n c^n : n, m > 0 \}$.

Hint:

$G = (\{S, B, C\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow aSBC$

2. $S \rightarrow aBC$

3. $aB \rightarrow ab$

4. $bB \rightarrow bb$

5. $bC \rightarrow bc$

6. $cC \rightarrow cc$

Write CSG to generate the language $L = \{ w : na(w) = nb(w), w \in \{a, b\}^* \}$.

Hint:

Grammar: $G = (\{S\}, \{a, b\}, P, S)$

Productions

1. $S \rightarrow a S b$

2. $S \rightarrow b S a$

3. $S \rightarrow S S$

4. $S \rightarrow \epsilon$

Write CSG to generate the language $L = \{ w : na(w) > nb(w), w \in \{a, b\}^* \}$.

Hint:

Grammar: $G = (\{S\}, \{a, b\}, P, S)$

Productions

1. $S \rightarrow a S b$

2. $S \rightarrow a S$

3. $S \rightarrow S S$

4. $S \rightarrow a$

Write CSG to generate the language $L = \{ w : na(w) < nb(w), w \in \{a, b\}^* \}$.

Hint:

Grammar: $G = (\{S\}, \{a, b\}, P, S)$

Productions

1. $S \rightarrow a S b$

2. $S \rightarrow b S$

3. $S \rightarrow S S$

4. $S \rightarrow b$

Write CSG to generate the language $L = \{ w : na(w) \neq nb(w), w \in \{a, b\}^* \}$.

Hint:

$G = (\{S, S1, S2\}, \{a, b\}, P, S)$

Productions

1. $S \rightarrow S_1$

2. $S \rightarrow S_2$

For $n_a > n_b$ (S_1):

3. $S_1 \rightarrow a S_1 b$

4. $S_1 \rightarrow a S_1$

5. $S_1 \rightarrow S_1 S_1$

6. $S_1 \rightarrow a$

For $n_a < n_b$ (S_2):

7. $S_2 \rightarrow a S_2 b$

8. $S_2 \rightarrow b S_2$

9. $S_2 \rightarrow S_2 S_2$

10. $S_2 \rightarrow b$

Write CSG to generate the language $L = \{ x \in \{a, b, c\}^* \mid N_a(x) = N_b(x) = N_c(x) \}$.

Hint:

$G = (\{S\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow aSbc$

2. $S \rightarrow bSac$

3. $S \rightarrow cSab$

4. $S \rightarrow abc$

Write CSG to generate the language $L = \{ x \in \{a, b, c\}^* \mid N_a(x) < N_b(x) \}$.

Hint:

$G = (\{S\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow aSb$

2. $S \rightarrow bS$

3. $S \rightarrow cS$

4. $S \rightarrow b$

Write CSG to generate the language $L = \{ x \in \{a, b, c\}^* \mid N_a(x) < N_c(x) \}$.

Hint:

$G = (\{S\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow aSc$

2. $S \rightarrow bS$

3. $S \rightarrow c$

Write CSG to generate the language $L = \{ x \in \{a, b, c\}^* \mid N_b(x) < N_c(x) \}$.

Hint:

$G = (\{S\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow bSc$

2. $S \rightarrow aS$

3. $S \rightarrow c$

Write CSG to generate the language $L = \{ x \in \{a, b, c\}^* \mid N_a(x) < N_b(x) < N_c(x) \}$.

Hint:

$G = (\{S\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow aSbc$

2. $S \rightarrow bSc$

3. $S \rightarrow c$

Write CSG to generate the language $L = \{ a^i b^j c^k : i < j \}$.

Hint:

$G = (\{S, A\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow aSb$

2. $S \rightarrow bS$

3. $S \rightarrow Ac$

4. $A \rightarrow cA$

5. $A \rightarrow c$

Write CSG to generate the language $L = \{ a^i b^j c^k : i < k \}$.

Hint:

$G = (\{S\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow aSc$

2. $S \rightarrow bS$

3. $S \rightarrow cc$

Write CSG to generate the language $L = \{ a^i b^j c^k : j < k \}$.

Hint:

$G = (\{S\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow bSc$

2. $S \rightarrow aS$

3. $S \rightarrow cc$

Write CSG to generate the language $L = \{ a^i b^j c^k : i < j \text{ and } i < k \}$.

Hint:

$G = (\{S\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow aSbc$

2. $S \rightarrow bc$

Write CSG to generate the language $L = \{ a^i b^j c^i d^i \mid i \geq 0 \}$.

Hint:

$G = (\{S, B, C, D\}, \{a, b, c, d\}, P, S)$

Productions

1. $S \rightarrow aSBCD$

2. $S \rightarrow aBCD$

3. $bB \rightarrow bb$
4. $cC \rightarrow cc$
5. $dD \rightarrow dd$
6. $aB \rightarrow ab$
7. $bC \rightarrow bc$
8. $cD \rightarrow cd$

Write CSG to generate the language $L = \{a^i b^j c^k d^l \mid i=j \text{ and } k=l \geq 0\}$.

Hint:

$G = (\{S, T\}, \{a, b, c, d\}, P, S)$

Productions

1. $S \rightarrow aSb$
2. $S \rightarrow T$
3. $T \rightarrow cTd$
4. $T \rightarrow \epsilon$

Write CSG to generate the language $L = \{w \mid w \in \{a, b, c\}^* \text{ and } w \text{ is a palindrome}\}$.

Hint:

$G = (\{S\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow aSa$
2. $S \rightarrow bSb$
3. $S \rightarrow cSc$
4. $S \rightarrow a$
5. $S \rightarrow b$
6. $S \rightarrow c$
7. $S \rightarrow \epsilon$

Write CSG to generate the language $L = \{ww^r \mid w \in \{a, b, c\}^*\}$.

Hint:

$G = (\{S\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow aSa$
2. $S \rightarrow bSb$
3. $S \rightarrow cSc$
4. $S \rightarrow \epsilon$

Write CSG to generate the language $L = \{ww \mid w \in \{a, b\}^*\}$.

Hint:

$G = (\{S, A, B\}, \{a, b\}, P, S)$

Productions

1. $S \rightarrow aSa$
2. $S \rightarrow bSb$
3. $S \rightarrow A$
4. $A \rightarrow aA$
5. $A \rightarrow bA$
6. $A \rightarrow \epsilon$

Write CSG to generate the language $L = \{ww \mid w \in \{a, b, c\}^*\}$.

Hint:

$G = (\{S\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow aSa$

2. $S \rightarrow bSb$

3. $S \rightarrow cSc$

4. $S \rightarrow \epsilon$

Write CSG to generate the language $L = \{ww^r \mid w \in \{a, b\}^*\}$.

Hint:

$G = (\{S\}, \{a, b\}, P, S)$

Productions

1. $S \rightarrow aSa$

2. $S \rightarrow bSb$

3. $S \rightarrow aa$

4. $S \rightarrow bb$

5. $S \rightarrow \epsilon$

Write CSG to generate the language $L = \{wcw \mid w \in \{a, b\}^*\}$.

Hint:

$G = (\{S\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow aSa$

2. $S \rightarrow bSb$

3. $S \rightarrow c$

Write CSG to generate the language $L = \{wcw^r \mid w \in \{a, b\}^*\}$.

Hint:

Grammar: $G = (\{S\}, \{a, b, c\}, P, S)$

Productions

1. $S \rightarrow aSa$

2. $S \rightarrow bSb$

3. $S \rightarrow aca$

4. $S \rightarrow bcb$

5. $S \rightarrow c$

Write CSG to generate the language $L = \{xxx \mid x \in \{a, b\}^*\}$.

Hint:

$G = (\{S\}, \{a, b\}, P, S)$

Productions

1. $S \rightarrow aSaaa$

2. $S \rightarrow bSbbb$

3. $S \rightarrow \epsilon$

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Context Sensitive Grammar (CSG):

The CSG is defined as $G=(V, T, P, S)$, where,

V: Non terminals or variables.

T: Input symbols.

S: Starting symbol.

P: Production rule. And are of the form

$$\alpha A \beta \rightarrow \alpha \gamma \beta,$$

where $A \in V$,

$$\alpha, \beta \in (V \cup T)^*$$

$$\gamma \in (V \cup T)^+$$

The rule is non-contracting, i.e., $|\alpha \gamma \beta| \geq |\alpha A \beta|$

CHEAT SHEET: CSG

1: First identify the constraint type in the language. What is the language checking?

There are only a few common patterns:

- Equal counts: $a^n b^n$, $a^n b^n c^n$, $N_a = N_b = N_c$
- Multiple relations: $a^{2n} b^n$, $a^n b^{2n}$, $a^{3n} b^n$
- Inequality relations: $N_a > N_b$, $i < j < k$
- Structural symmetry: palindromes, ww , ww^r , wcw

Once you classify the language, grammar design becomes mechanical.

2: Use the “generate–match–convert” template.

Step A: Generate symbols in linked groups

Step B: Enforce equality or inequality using markers (variables)

Step C: Convert variables into terminals at the end

Example template:

$$S \rightarrow aSBC$$

$$S \rightarrow aBC$$

$$aB \rightarrow ab$$

$$bB \rightarrow bb$$

$$bC \rightarrow bc$$

$$cC \rightarrow cc$$

This single pattern solves:

- $a^n b^n$
- $a^n b^n c^n$
- $a^k b^n$
- $a^n b^k$

3: For equal-count languages (languages like: $N_a = N_b = N_c$), use one of these two beginner-safe patterns:

Pattern 1 (Direct generation)

$$S \rightarrow aSbc$$

$S \rightarrow bSac$

$S \rightarrow cSab$

$S \rightarrow abc$

Pattern 2 (Generate then permute)

$S \rightarrow ABCS$

$S \rightarrow ABC$

$AB \leftrightarrow BA$

$AC \leftrightarrow CA$

$BC \leftrightarrow CB$

$A \rightarrow a$

$B \rightarrow b$

$C \rightarrow c$

4: For palindromes / symmetry languages, use mirror rules (languages like: palindrome, ww^r , wcw), use this structure:

$S \rightarrow aSa$

$S \rightarrow bSb$

$S \rightarrow cSc$

$S \rightarrow \text{middle symbol or } \epsilon$

5: For inequality languages, split into cases (languages like: $N_a \neq N_b$, $N_a < N_b$, $i < j < k$), Do NOT try to do it in one grammar rule.

Instead:

$S \rightarrow S_1 \mid S_2$

S_1 handles $N_a > N_b$

S_2 handles $N_a < N_b$

This reduces errors dramatically.

6. What NOT to do !

Do NOT:

- invent new symbols without purpose
- mix multiple constraints in one rule
- try to optimize grammar
- use fancy permutations early

NOTE:

Context Sensitive Grammar (CSG) Rule Constraint

A grammar is context sensitive if all productions are non-contracting, meaning:

For any rule $\alpha \rightarrow \beta$, we must have $|\beta| \geq |\alpha|$

$|\alpha|$ = length of LHS

$|\beta|$ = length of RHS

Exception: $S \rightarrow \epsilon$ is allowed only if S does not appear on any RHS (i.e., only as the start symbol)

So technically:

1. All rules must not shrink the string (no contracting rules)

2. $S \rightarrow \epsilon$ is allowed only if ϵ is in the language and S does not appear on any RHS

When is $S \rightarrow \epsilon$ allowed?

Allowed only if:

1. S is the start symbol
2. S does not appear on the RHS of any other rule
3. ϵ is in the language

C

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Improvement CIE: Syllabus

Unit –IV

Context Sensitive Languages (CSL) and Linear Bounded Automata (LBA), Turing Machines (TM): Definitions and Examples, TM as a Language Acceptor, Computing Partial Functions with Turing Machine, Variations of Turing Machines, Combining Turing Machines, Non Deterministic TM, Universal TM.

Unit –V

Recursively Enumerable Languages (REL) and Recursive Languages. Properties of REL and Recursive Languages. More General Grammars: Context Sensitive Grammar and Unrestricted Grammar, Chomsky Hierarchy, Not all languages are Recursively Enumerable, Unsolvable Problem, Reducing One problem to another, The halting problem of TM, Post's Correspondence Problem (PCP), Time and Space Complexity of TM

C

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Classwork: Turing Machine (easy-Level)

1. Design a Turing Machine to accept strings with even number of 1s over $\{0,1\}$
2. Design a Turing Machine to accept strings containing at least one a over $\{a,b\}$
3. Design a Turing Machine to accept strings of the form 0^*
4. Design a Turing Machine to accept strings that contain exactly one 1.
5. Design a Turing Machine to accept strings that start and end with the same symbol over $\{a,b\}$
6. Design a Turing Machine to accept palindromes of length 1 over $\{a,b\}$
7. Design a Turing Machine to accept strings of the form $a^n b^m$, where $n, m \geq 0$
8. Design a Turing Machine to accept strings with equal number of a s and b s where all a come before b s.
9. Design a Turing Machine to replace all 0s with 1s in the input string.
10. Design a Turing Machine that copies the input string once.
11. Design a Turing Machine that changes all a s to b s.
12. Design a Turing Machine that accepts only if the input length is even.